

Yuyao Wu

📍 Hong Kong, China

✉ Lexie.Wucq@gmail.com



QUALIFICATIONS

- National Computer Rank Examination Level 4
- Mandarin Level II, Grade A
- English Proficiency: IELTS 7.5, CET 6-651, CET 4-649

EDUCATION

Bachelor in Telecommunications Engineering

2021-2025

Huazhong University of Science and Technology, Wuhan, China

- GPA: 91.1/100
- Overall performance ranking: **2/116**
- Related coursework: Fundamentals of Wireless Communications(100), Circuit Testing Lab(99), Calculus(96), Software Radio(93), Digital Image Processing(93), Probability Theory and Mathematical Statistics(93), Linear Algebra(91)

RESEARCH PROJECTS AND PUBLICATIONS

Projects

- ★ **Leader of A National Undergraduate Training Program for Innovation and Entrepreneurship**
- **Channel Sensing for Holographic Interference Surfaces (HIS):** Proposed a novel architecture integrating a phase-shifting-based interference suppression method to mitigate self-interference during the holographic recording process.
- **Multi-User Channel Segmentation Algorithm:** Developed a segmentation algorithm for HIS leveraging extended Prony's method to accurately estimate channel state information (CSI) in multi-user scenarios by utilizing spatial characteristics of uniform planar array (UPA) channels.
- **Low-Rank Channel Estimation:** Designed a channel estimation algorithm for HIS exploiting the inherent low-rank property of wireless channels.
- **Noise-induced Estimation Error Elimination:** Designed a multiple temporal sampling method to statistically mitigate the estimation error.
- **Theoretical Validation and Simulation:** Provided rigorous theoretical proofs and comprehensive simulations to validate the proposed theorems, lemmas, and algorithms.

Journal Articles

- J. Huang, **Y. Wu**, H. Yin, Y. Zhang and R. Zhang, "Channel Sensing for Holographic Interference Surfaces Based on the Principle of Interferometry," in IEEE Transactions on Wireless Communications, vol. 23, no. 7, pp. 7953-7966, July 2024, doi: 10.1109/TWC.2023.3347257.
- **Y. Wu**, H. Yin, J. Huang, "Channel Estimation with Partial Hologram Sampling for Holographic Interference Surfaces", in preparation for its submission to TCOM.

Conference Articles

- J. Huang, **Y. Wu**, H. Yin, Y. Zhang and R. Zhang, "Channel Estimation based on Interference Principle for Holographic Interference Surfaces," 2023 IEEE Globecom Workshops (GC Wkshps), Kuala Lumpur, Malaysia, 2023, pp. 1063-1068, doi: 10.1109/GCWkshps58843.2023.10464753.
- **Y. Wu**, H. Yin, J. Huang, "A Channel Sensing Method with Partial Hologram for Holographic Interference Surfaces", accepted by 2024 IEEE 24th International Conference on Communication Technology (IEEE ICCT 2024).

HONORS & AWARDS

Honors-PhD

- **A recipient of 2025 Hong Kong PhD Fellowship Scheme (HKPFS)**

Honors-Bachelor

- **Pacemaker to Merit Student of Huazhong University of Science and Technology**
Statement: It represents the highest honor for undergraduates at HUST, awarded annually to the **20** most outstanding students, who constitute less than **0.01%** of the entire student body.
- **2021 - 2022, 2023 - 2024 China National Scholarship**
- **2022 Outstanding Undergraduates in Term of Academic Performance**
- Scholarship for Scientific and Technological Innovation of Huazhong University of Science and Technology
- 2024 Representative of Senior Students for Huazhong University of Science and Technology, School of Electronic Information and Communications
- Outstanding Undergraduate Admission Volunteer
- University-Level Outstanding Practice Team (Team Leader) & Individual

Awards

- 2021 China Collegiate Programming Contest for Girls - **National Gold Medal**
- China International College Student's Innovation Competition 2023 - **National Gold Medal**
- 2022 National English Competition for College Students(Band C) - Second Prize
- 2023 China Undergraduate Mathematical Contest in Modeling - Third prize
- 5th Sports Skills (Basketball) Competition for PE Classes at HUST - Second Prize
- Women's Team Event in 2021 HUST Table Tennis League - Fifth Place

STUDENT WORK EXPERIENCE

- | | |
|---|-----------------------|
| • Class monitor | Sept 2021 – July 2022 |
| • Head of the College Sports Department | Dec 2021 – Aug 2022 |
| • Assistant to Student Affairs Department | Sept 2022 – June 2025 |
| • Secretary of Youth League branch | Sept 2022 – June 2025 |
| • Department Employment Assitant | May 2024 – June 2025 |

EXTRACURRICULAR ACTIVITIES AND VOLUNTARY WORK

- | | |
|---|-----------------------|
| • Summer School in INP-ENSEEIH of Toulouse, France – Team Leader | July 2024 |
| • Internet Enterprise Visit – Team Leader | Jan 2024 |
| • Communication Enterprise Visit | Nov 2023 |
| • Field Research – Team Leader | Aug 2024 |
| • Women's Basketball Team Member of School | Sept 2021 - June 2025 |